Computational Spectral Investigation of the Sloan Digital Sky Survey Star SDSS 2942-54521-0124 and Determination of its Stellar Parameters

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Abstract

Spectroscopy remains one of the most powerful observational techniques in astrophysics, enabling astronomers to understand the physical properties of stars from their emitted radiation. In this work, the optical spectrum of the star SDSS 2942-54521-0124 was retrieved from the Sloan Digital Sky Survey (SDSS) archive and analyzed using Python-based computational tools. The spectral dataset stored in Flexible Image Transport System (FITS) format was processed to extract wavelength and flux information and reconstruct the stellar spectrum. The resulting spectral distribution shows a continuum radiation profile with several absorption features corresponding to atomic transitions. Hydrogen Balmer lines and absorption features associated with sodium, magnesium, and calcium were identified in the spectrum. Using the characteristics of the spectral distribution and the position of the continuum peak, approximate physical parameters of the star were inferred.

Keywords: Stellar Spectroscopy, Computational Astrophysics, Stellar Physical Parameters, Sloan Digital Sky Survey (SDSS).

1 Introduction

Stellar spectroscopy provides one of the most direct methods for understanding the physical nature of stars. Since stars cannot be physically sampled, the only information available to observers is the electromagnetic radiation emitted from their surfaces. When this radiation is dispersed into its component wavelengths, the resulting spectrum contains detailed information regarding temperature, chemical composition, and kinematic properties of the stellar atmosphere [2].

The radiation emitted by a star is often approximated by thermal radiation. The distribution of emitted radiation as a function of wavelength can be described by Planck's radiation law

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda kT)} - 1} \tag{1}$$

where B_λ represents the spectral radiance, T is the temperature, h is Planck's constant, c is the speed of light, and k is Boltzmann's constant [3].

Large astronomical surveys have greatly expanded the availability of spectroscopic data. One of the most influential surveys is the Sloan Digital Sky Survey (SDSS), which has obtained spectroscopic observations of millions of celestial objects across large portions of the sky [1].

2 Data Source

The spectral dataset analyzed in this study was obtained from the Sloan Digital Sky Survey archive. The star is identified by the observation parameters:

Plate 2942, MJD 54521, Fiber 0124

The spectral data are stored in Flexible Image Transport System (FITS) format. The wavelength values were converted using

$$\lambda = 10^{\log \lambda} \tag{2}$$

which transforms the logarithmic wavelength scale into linear wavelength units.

3 Methodology

The spectral analysis was performed using a Python computational environment. The FITS file was opened using the Astropy library, which provides tools for handling astronomical data formats. The wavelength and flux arrays were extracted and used to reconstruct the stellar spectrum.

The spectrum was plotted using Matplotlib, producing a flux versus wavelength distribution.

4 Spectral Line Identification

The stellar spectrum exhibits several absorption features corresponding to atomic transitions in the stellar atmosphere.

4.1 Hydrogen Balmer Lines

Hydrogen Balmer lines arise when electrons transition between higher energy levels and the $n = 2$ energy state. The wavelengths of the Balmer series are given by the Rydberg equation

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \tag{3}$$

where $R_H = 1.097 \times 10^7\, m^{-1}$.

Prominent Balmer lines observed include

$$\begin{aligned} \mathrm{H}\alpha &: 6563\ \text{Å} \\ \mathrm{H}\beta &: 4861\ \text{Å} \\ \mathrm{H}\gamma &: 4341\ \text{Å} \end{aligned}$$

4.2 Sodium, Magnesium and Calcium

Additional absorption features correspond to heavier elements:

$$\begin{aligned} \text{Na D} &: 5890\ \text{Å} \\ \text{Mg I} &: 5175\ \text{Å} \\ \text{Ca II} &: 3933\ \text{Å} \end{aligned}$$

5 Radial Velocity Estimation

The radial motion of a star relative to the observer can be determined from the Doppler shift of spectral lines. The dimensionless redshift parameter z is defined as

$$z = \frac{\lambda_{obs} - \lambda_0}{\lambda_0} \tag{4}$$

where λ_{obs} represents the observed wavelength and λ_0 is the rest wavelength of the spectral line.

For non–relativistic velocities, the radial velocity of the star can be approximated by

$$v = zc \tag{5}$$

where c is the speed of light.

From the measured displacement of absorption features in the stellar spectrum, the redshift parameter is estimated to be of the order

$$z \approx 1.7 \times 10^{-5} \tag{6}$$

Substituting this value into the velocity relation gives

$$v = zc \approx (1.7 \times 10^{-5})(3 \times 10^5 \text{ km s}^{-1}) \tag{7}$$

$$v \approx 5 \text{ km s}^{-1} \tag{8}$$

The negative sign of the shift indicates that the star exhibits a small radial motion toward the Earth.

6 Surface Temperature Estimation

The effective surface temperature of the star can be approximated using Wien's displacement law

$$\lambda_{max} T = b \tag{9}$$

where $b = 2.897 \times 10^{-3}\, mK$.

If the spectral peak occurs near

$$\lambda_{max} \approx 5000\,\text{Å} \tag{10}$$

the approximate stellar temperature becomes

$$T \approx 5800\, K \tag{11}$$

7 Identified Spectral Lines

Element	Wavelength (Å)	Feature
Hydrogen (Hα)	6563	Balmer line
Hydrogen (Hβ)	4861	Balmer line
Sodium (Na D)	5890	Sodium doublet
Magnesium (Mg I)	5175	Magnesium absorption
Calcium (Ca II)	3933	Calcium absorption

8 Results

The reconstructed stellar spectrum displays a continuum radiation distribution with several absorption features, as shown in Figure 1.

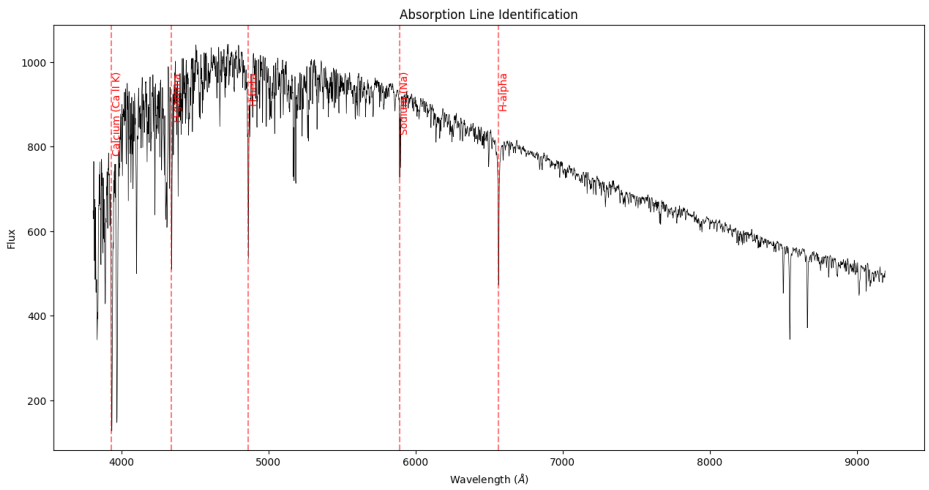

Figure 1: Computed stellar spectrum of SDSS 2942-54521-0124 showing the continuum distribution and absorption features corresponding to hydrogen (Hα, Hβ), sodium (Na D), magnesium (Mg I), and calcium (Ca II).

Hydrogen Balmer lines are among the most prominent features in the spectrum.

The original observational spectrum obtained from the Sloan Digital Sky Survey is shown in Figure 2. This spectrum represents the primary data set used for the computational analysis performed in this study.

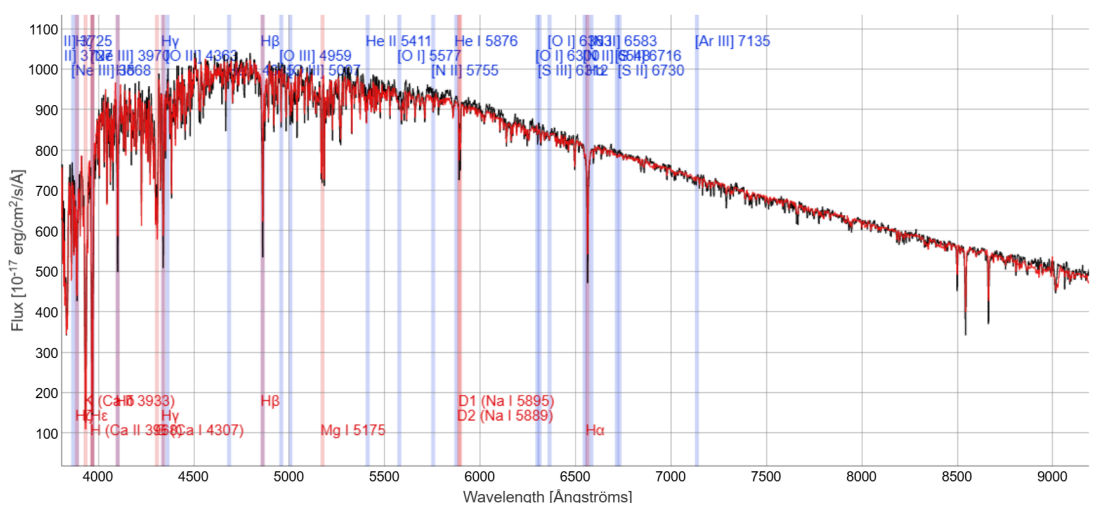

Figure 2: Optical spectrum of the star SDSS 2942-54521-0124 obtained from the Sloan Digital Sky Survey database, showing flux as a function of wavelength with labeled spectral transitions.

The comparison between the computed spectrum and the observational data-set confirms the presence of several absorption features corresponding to hydrogen, sodium, magnesium, and calcium.

9 Discussion

The identification of hydrogen absorption lines indicates hydrogen-rich stellar material, while the presence of heavier elements such as sodium, magnesium, and calcium confirms the complexity of stellar atmospheres.

10 Conclusion

The optical spectrum of the star identified by SDSS observation Plate 2942, MJD 54521, Fiber 0124 was analyzed using computational tools. Hydrogen Balmer lines and additional absorption features associated with sodium, magnesium, and calcium were detected.

References

[1] York, D. G., Adelman, J., Anderson, J. E., et al. (2000). The Sloan Digital Sky Survey: Technical Summary. *The Astronomical Journal*, 120, 1579–1587.

[2] Gray, D. F. (2005). *The Observation and Analysis of Stellar Photospheres*. Cambridge University Press.

[3] Carroll, B. W., & Ostlie, D. A. (2017). *An Introduction to Modern Astrophysics*. Cambridge University Press.

[4] Rybicki, G. B., & Lightman, A. P. (1979). *Radiative Processes in Astrophysics*. Wiley.

[5] Rutten, R. J. (2003). *Radiative Transfer in Stellar Atmospheres*. Utrecht University Lecture Notes.

[6] Morgan, W. W., Keenan, P. C., & Kellman, E. (1943). *An Atlas of Stellar Spectra*. University of Chicago Press.

[7] Abazajian, K., et al. (2009). The Seventh Data Release of the Sloan Digital Sky Survey. *Astrophysical Journal Supplement Series*.

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